RULES FROM THE GAME – Version 1.14

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#1 - AIR UNITS

Air Units

Each counter stands for 4 planes

Air to air factor: ability to attack other planes

Fighters

Air to Ground attack factor: ability to hit ground targets - () = no bombs, strafe only

In Blue: Can operate on carriers

Level Bombers

Search radius in hexes
Means this unit is currently searching up to this radius around its airbase

Dive Bombers

Ruggedness: ability to sustain damage

Torpedo Bombers

Air to Sea factor: ability to hit naval targets

Surface search radar

In Green: Strategic Bomber

In White/Orange: Large/Small seaplanes. Managed by the program

Remaining endurance
Number of HEXES it can fly or number of PHASES it can stay aloft before running out of fuel
1 pt is spent per PHASE aloft or 2 pts if moved by 2 HEXES this phase

Payload

TORPEDO — Carried by certain plane models, can only hit ships, lower chances to hit but more lethal

BOMBS — Carried by almost all plane models, attacks both ships and bases, highly effective against flight decks but less against ship structures
A fighter equipped with bomb is called a fighter-bomber

DROP TANKS — Usually the default payload
They increase the endurance at the expense of the offensive payload

CAP — Fighters in CAP do not carry drop tanks and thus have a reduced endurance

TRANSFER — No armament at all
The best way to rebase from one friendly airbase to another

When you assign a mission to air units, the game selects for you the best payload fitting the mission depending on the distance and the target
You may still have the option to send your fighters as bombers or escort

Air units inventory

	CAP	Drop Tank	Bombs	Torpedo	Transfer	Scout
Boston						
Hudson						
Claude						
Zero						
Hamp						
Rufe						
Anson						
Flying Fortress						
Liberator						
Mitchell						
Marauder						
Jean						
Kate						
Jill						

Beaufighter						
Beaufort						
Boomerang						
Val						
Judy						
Jake						
Pete						
Buffalo						
Wildcat						
Corsair						
Hellcat						
Nell						
Betty						
Mavis						

Emily					
Sally					
Oscar					
Nick					
Lily					
Helen					
Sonia					
Tony					
Lightning					
Airacobra					
Kittyhawk					
Thunderbolt					
Black Cat					
Helldiver					

Vindicator

Dauntless

Seagull

Spitfire

Devastator

Avenger

Albacore

Blenheim

Fulmar

Hurricane

Martlet

Airfields

They are 3 types of airfields:

Operates ONLY Fighters

Operates any plane
BUT Strategic Bombers

Operates ANY planes

Airfield availability, runways and type depends on the scenario

Endurance and Ammunition

When the remaining endurance reaches zero, air units may still fly on their fuel reserve, up to 4 hexes.
The effects are the following

No effect

10% probability of
crash at landing

30% probability of
crash at landing

50% probability of
crash at landing

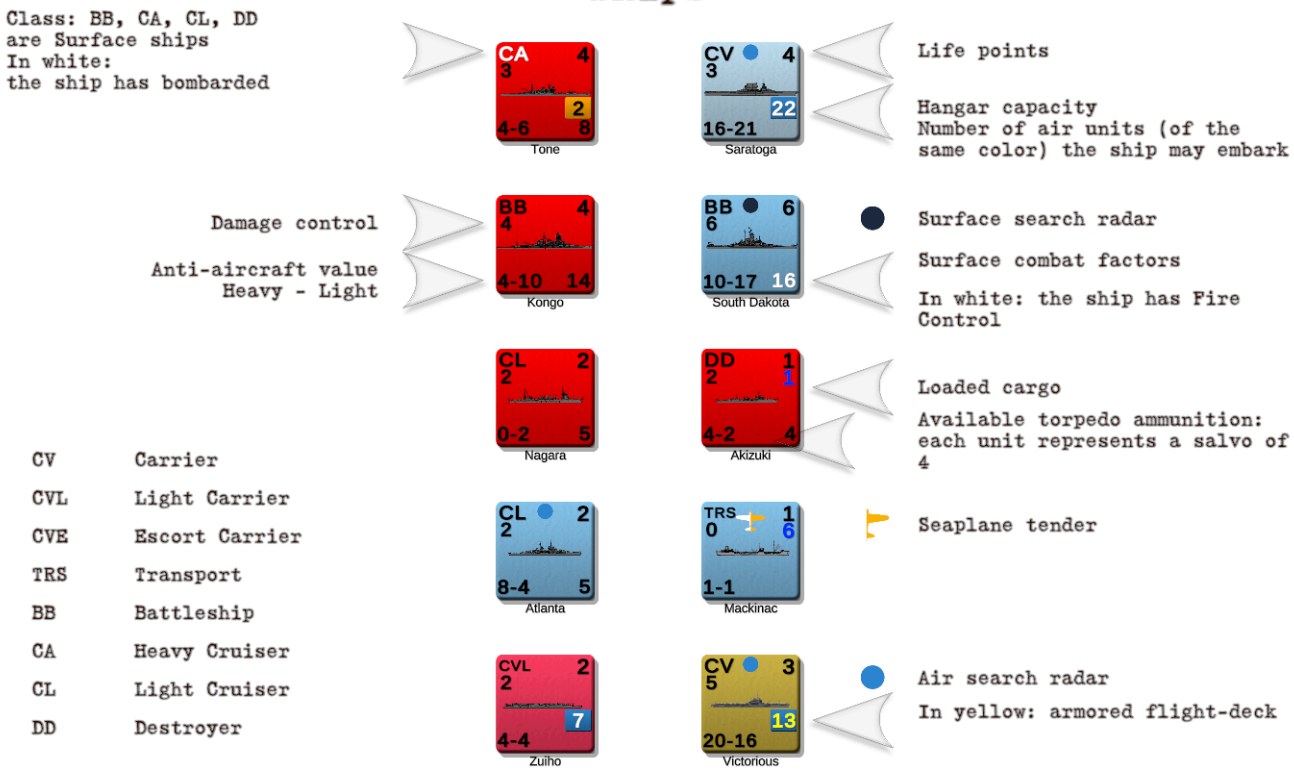
70% probability of
crash at landing

Eliminated
below -4

Because of ammunition depletion, each battle fought reduces all combat factors to zero
Exception: a CAP fighter has a 50% chance to keep its ammunition
Depleted units need to land on a friendly airbase in order to be able to fly more missions at full potential

#2 – SHIPS AND FORCES

Ships

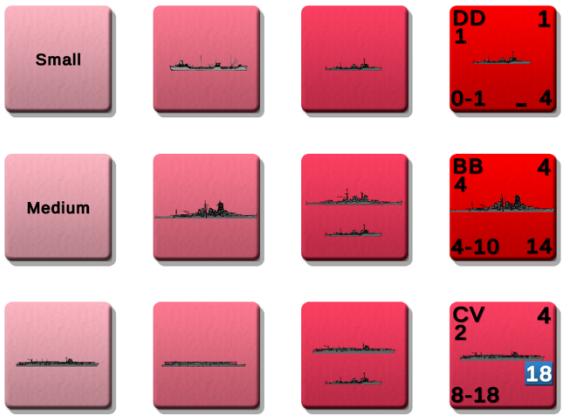


Counter back

Task Forces and Intelligence

Ships move and fight together in Task Forces (TF)

The more Intelligence you gain on a Japanese TF, the more details you get on its composition



Task Forces and Intelligence

Undetected TF are not visible on map and
can't be the target of any airstrike

TF are usually detected by air search. Each time a TF is detected, it appears on map and its intelligence may increase by 1 level

Indicates the last observed direction taken by the detected TF

If an enemy task force has not been detected after 3 TURNS (1 at night), it disappears from the map

Enemy TF are displayed in their last observed hex
They may move undetected and hence may not exactly be where you see them on map

If an airstrike or TF arrives at a hex where an enemy Task Force was spotted but the enemy Task Force is no longer there. The marker will be removed from the map

Detach a Task Force

A task group may be detached in any hex if:

- There are fewer than 20 friendly Task Forces in play (retiring TF and submarines do not count towards this limit)
- There are at least 4 ships in the current TF

When you detach a carrier from a TF containing several carriers, any airborne airstrike of this TF may be split

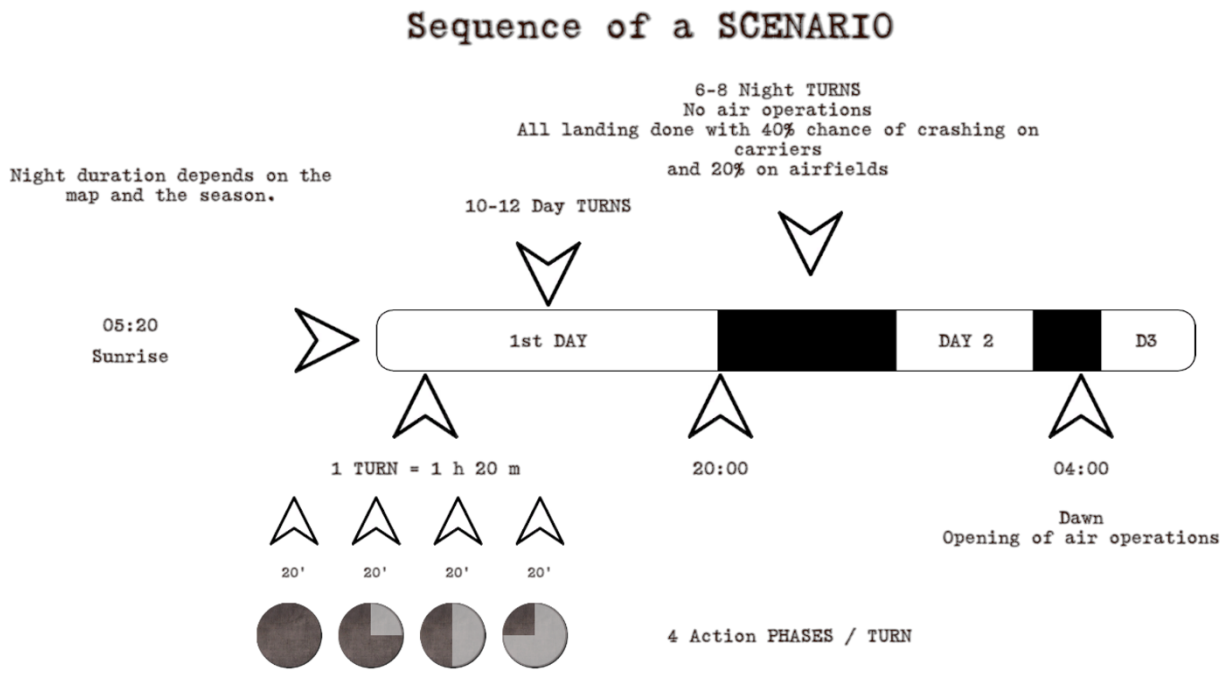
Merge Task Forces

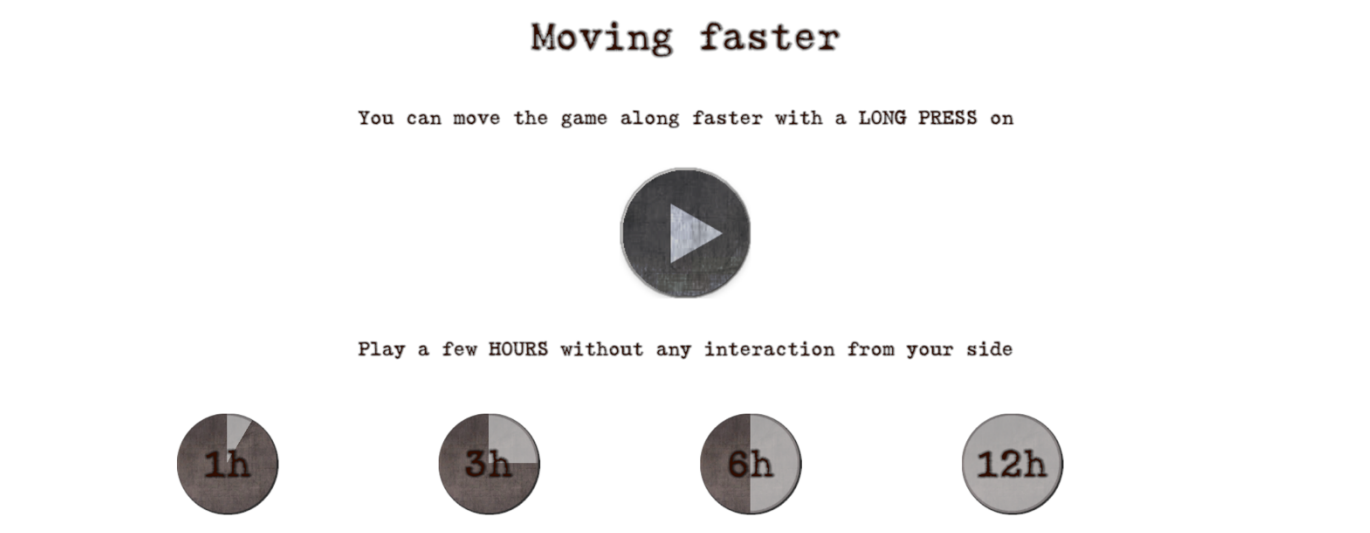
You may freely merge TF

- which are not retiring
- which are in the same hex
- without disorder point
- as long as a TF does not contain more than 12 ships

IMPORTANT: A merged TF is not allowed to move this TURN

#3 – SEQUENCE OF PLAY





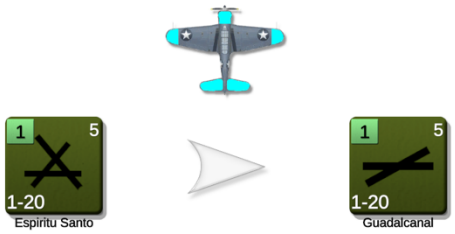
#4 - AIR OPERATIONS

VIDEO TUTORIAL ON YOUTUBE : https://www.youtube.com/watch?v=pJBbHqAlNso

You have 2 tutorials at your disposal to learn the basics of air operation management. Is is very important to follow them before starting a real game

Transfer

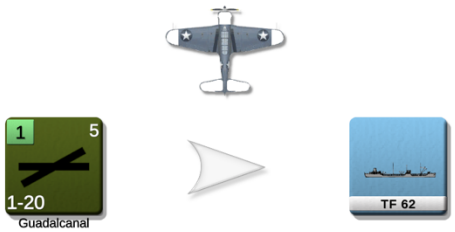
You may rebase air units between 2 allied bases
Select the destination base from the STRIKE box of the start base



Transfer missions work like airstrikes and the destination base is always found
Transfer missions are moving only 1 hex per PHASE

Long-range CAP

Units already on CAP may temporarily protect another airfield or TF
Select the target to protect and the number of fighters to allocate to this mission from the CAP box of the start base



Long-range CAP units will move towards the target and act as CAP as long as it remains above it

Fighters will return to base if out of ammunition or if forced by remaining endurance or daylight

After completing their mission, air units will be automatically reallocated to CAP, except for fuel-limited airfields

Repetitive Missions

Select a stack of CAP/searching air units to make their mission repetitive

CAP mission are repetitive by default, except on airfields with limited fuel supply

The repetition stops:

- By a new selection of the air units
- When cancelling the mission by moving the air units to the hangar or to another kind of mission
- At night

Automation

The game will manage automatically without any action from your side

- The preparation and the launch of missions
  The fighters will show up in the CAP box once aloft

- The relentless cycle landing-refuel-launch of CAP fighters
  This will ensure a constant cover over the airbase
  Emphasized by the blue circular arrow on air units

If you want a mission to have precedence over others, set it as top priority
Select on the flag in the Air op screen to do so

To cancel all or part of a mission, drag the units to the HANGAR
You may also reallocate them to a different mission

Carrier-based air units will always attempt to land on their original carrier first

Limitations

The following factors limit air operations

- A carrier cannot land and launch air units within the same PHASE
- With several missions at the same time or with many units, the resulting traffic on deck/runway, elevators or in the hangar will slow down operations
- A single-wave air strike will take more time to mount
- A carrier or an airfield may take damage slowing down or hindering operations
- Air units may not land or launch from a carrier on fire, crippled or with a damaged flight deck
- An airfield without any operational runway may not launch aircraft but may still land them with a risk of crash

On average, each carrier or runway may operate 8 air units per PHASE
Lexington, Saratoga, CVL and CVE have fewer elevators and usually manage 6 units per PHASE

Some airfields have a limited fuel supply
Each launched unit burns fuel points: strategic bombers 3, bombers 2, fighters 1, each pair of CAP fighter launched the same PHASE 1

Traffic

The flags above the deck/runway are reflecting the state of the traffic on air operations

Operations proceed optimally

Traffic-jam, units are somehow hindering each other. The number indicates the cumulated delay in multiple of 20 minutes. Ex: 2 may mean that a unit has been delayed by 40 min. or 2 units may be delayed by 20 min. Play with priority or reduce the number of aircraft trying to launch

No operation is allowed

Ready on deck/runway

OPTIONAL. You might want to prepare and keep as reserve on the deck a few air units for a quick reaction or opportunity
Just move them on DECK/RUNWAY box and select a payload
Such air units will be very quick to take off

Maintaining air units on the deck/runway may reduce the pace of the air operations especially for carriers. Choose well the payload because changing it once on the deck, will take a lot of time

#5 - AIR SEARCH

VIDEO TUTORIAL ON YOUTUBE : https://www.youtube.com/watch?v=PiGg4qyPJAg

Search for enemy Task Forces

There are several ways to search for enemy TF

You may send search units from airfields or carriers

US surface TF and all Japanese TF will send seaplanes from capable cruisers. Managed by the program

Seaplanes are also launched from seaplane bases which are not visible to the player and again managed by the program

Airstrikes will attempt to detect enemy TF when arriving at the target area but also while en-route

Each TF adjacent to enemy Task Force or airbase may be detected

Coastwatchers may detect TF and airstrikes in their hex
They are not visible on map. Pro-US are located in all large Solomon islands and in New Guinea. Pro-Japanese are in New Britain

Unloading transport TF are immediately detected during daylight
A TF performing naval bombardment is automatically detected

Air Search formula

The chance is determined by this formula

$$4 \times N \times 40 * (M + 1) * W / (30 \times (a + D \times A / 60))$$

- D = distance from the searching airbase to the searched TF
  +1 for every 3 hexes, ex: 9 counted as 12
- N = number of searching units able to reach the TF. Non-scouts units count for half
- A = arc of the search pattern
- W = weather effect in the TF hex (0.2 Storm, 0.35 Rain, 0.5 Overcast, 0.8 Clear before July 42, Otherwise 1)
- a = 1 or 0 if 360° arc
- M = Sum of modifiers:
- +0.01 per ship above the first
- +0.10 if the TF contains at least one BB/CV
- +0.50 if any ship has an oil leak or a major fire

Search radius

Each searching air unit covers an arc circle shape centered on its airbase
Its search radius is displayed on top right
Scout planes have a horizon of 1 hex from their search radius depending on their journey

This TBF has just started its mission and may search all hexes at 5 or 6 hexes around its carrier
Planes look on the side and in front of them
The search radius increases by 1 each PHASE

This TBF has reached its turning point and may search all hexes at 8, 9 or 10 hexes around its carrier
Here planes look in all directions

This TBF is returning and may search all hexes at 5 or 4 hexes around its carrier
The search radius decreases by 1 each PHASE

Fighters may not search as they lack required radio equipment and extra crew

The search radius is limited by the remaining endurance, nightfall, and the maximum search radius you may define

Detection of scouts

When an air unit detects a Task Force, it may be detected itself

A D10 is rolled. The scout is detected on a 6 or more

- +1 if CAP above the TF
- +2 if at least a US carrier with radar
- +1 if at least a ship with air radar
- -2 overcast
- -3 rain
- -4 light storm
- +1 if the scout was already shadowing this TF
- A natural 1 is always a failure

Without detection or CAP, the scout stays in the surroundings to shadow the TF
This TF will be automatically detected on the next PHASE

The plane symbol reminds the presence of a shadowing scout around the TF

Interception of scouts

Scouts may be intercepted by CAP provided they have been detected

A D10 is rolled

- +1/4 per fighter in CAP. Random rounding and maximum +2
- \+ air-to-air rating of the best fighter
- \- air-to-air rating of the scout
- \- scout ruggedness
- +2 US CAP
- -2 overcast
- -3 rain
- -4 light storm

Results

< 6	Chased away. The air unit resumes its patrol but may not shadow the Task Force 10% chance to transmit partial information on the TF, 20% no info transmitted at all
6-8	Abort. The scout returns to base and may not search until it lands 20% partial info, 30% no info
9-12	Damaged. Aborted mission and the air unit gets damaged 30% partial info, 40% no info
13+	Destroyed. The air unit gets immediately destroyed 40% partial info, 50% no info

Partial information: the TF position will be approximative

- The location error in hexes may be up to 1/4 (rounding down) of the distance from the scout's airbase
- The error must be smaller than the current error

Intelligence upgrade

When a Task Force is detected, its intelligence level may increase by 1

One D10 is rolled per current intelligence level
The level increases is the lowest die at least equals 6

- +2 if a scout is shadowing this TF
- +1 if the detection has been achieved by a TF equipped with surface radar
- +1 with clear weather if the detection was performed by an air unit
- -1 rain
- -2 light storm
- -3 if the detection has been done by a coastwatcher
- -2 at night
- Automatic for a TF which was not visible on map (level 0)

A TF vanishes from map and its level becomes zero

- After 4h after last detection at day
- After 1h20 after last detection at night

Detection of Task Forces from the ground

TF may be detected by adjacent enemy forces, airfields or coastwatchers

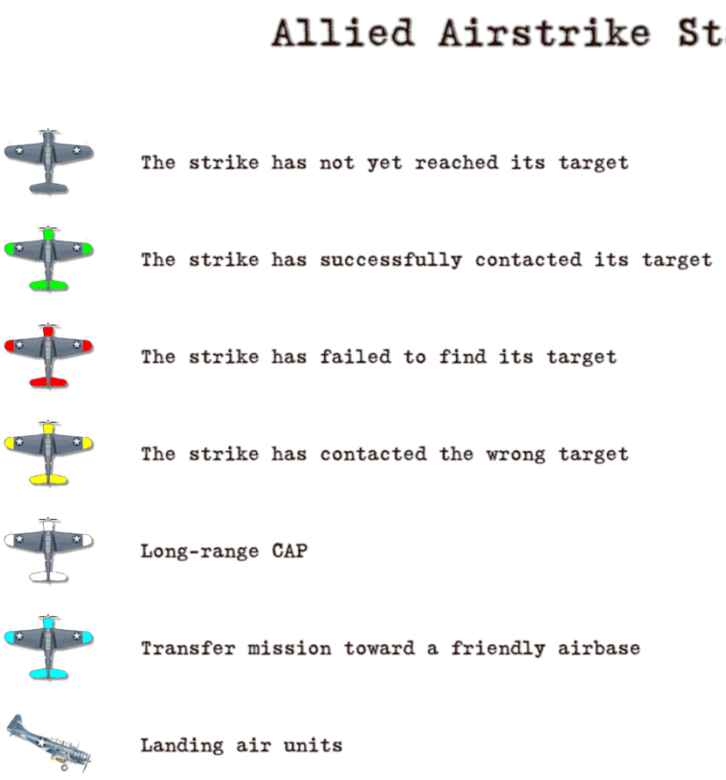
Distance	0	1
TF or airfield	60 %	30 %
Coastwatcher	40 %	

Modifiers

- -10 US coastwatchers in 1943
- +20 with surface radar
- -20/-10 at night US / Japanese
- +10 TF with seaplane at day or Japanese
- weather effect when not using radar

#6 - AIRSTRIKE

VIDEO TUTORIAL ON YOUTUBE : https://www.youtube.com/watch?v=mK3upia-8pg



Recall and Target change

Air units will abort and return to start base
The chance of success is equal to the radio efficiency
Automatic for transfers, airstrikes returning to base or long-range CAP
One attempt allowed per PHASE

Air units will change target or return base
One attempt allowed per PHASE

Radio efficiency

Radio transmission has an impact on:

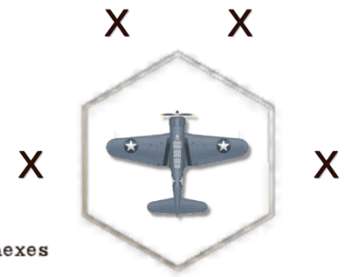
- The probability for an airstrike to be informed of the last position of its target
- The probability to change the target of an airstrike
- The probability to recall an airstrike

The efficiency is equal to 100% minus 10% (20% for the Japanese, doubled without any bomber) for each hex to the closest radio source

A source is a friendly Task Force, airfield

Detection of enemy Task Forces

An airstrike may detect enemy TF



- En-route
  Only in forward hexes (marked with a X)
- When reaching its target
  In all adjacent hexes

The detection chance is 200% in the airstrike hex, 100% in adjacent hexes
Modifiers:

- x Weather in TF hex (0.2 storm, 0.35 rain, 0.5 overcast, 1 clear)
- +1% per ship in the TF above the first
- +10% if the TF contains at least a BB/CV
- +50% if the TF has an oil leak
- Halved with single air unit. Halved without bomber
- x % of Japanese air units which will have at least 2 pts of endurance upon landing on CV with a minimum of 50% (planes needs fuel to find their carrier in absence of homing beacon)

Contacting the target

Each phase, each airstrike moves 2 hexes (1 for transfer, seaplanes, or with storm) closer to the last known position of its target
Each time the target is successfully searched, if the radio is efficient, then this position is updated

Air units will abort the airstrike if they have just enough endurance to return home. A little risk of 2 pts of endurance is granted against a target CV or when the target is detected in an adjacent hex

If the airstrike manages to reach its target, it attacks it
Airfields are automatically found because they don't move

If unsuccessful, the airstrike attacks an enemy detected Task Force in adjacent hexes

Otherwise, it will attempt to follow the last known direction taken by the TF to a maximum of 2 hexes

If nothing is found, then the airstrike returns to base without attacking

On its way back, it will attack any enemy TF detected and in the same hex

Contact

When target is successfully contacted by an air strike

- Then a Naval Air battle starts
- Target immediately becomes detected and visible on map

- Target Intelligence increases to level 4

Contact chances are good if

- The airstrike contains a lot of units
- The last known position of the target is frequently updated through an efficient search

Warning

Radar gives a tremendous advantage to the Allies
Each time an air strike moves within 4 hexes of an enemy airbase, it may be detected
Detected air strikes are visible on map and leave time for the defense to react

Distance	0	1	2	3	4
US Airfield with radar	90 %	70 %	50 %	30 %	10 %
US carriers with radar or Japanse airfield with radar	80 %	60 %	40 %	20 %	
US surface with radar, US airfield or Japanese carrier with radar	70 %	50 %	20 %		
Japanese surface ship with radar, Japanese airfield CAP, scout or friendly airstrike	60 %	30 %			
Ship without radar and Coastwatcher	30 %				

- +10 for the US in 43
- -20 against an air strike containing less than 4 air units
- +10 against an air strike containing more than 10 air units
- against an airstrike containing at least a 2 or 4-engines air unit
- A Coastwatcher in an enemy airfield hex use the row above
- weather effect when not using radar

Airstrike Intelligence

Enemy airstrikes are not fully detailed on map until they enter battle. Instead you get progressive information on their size and composition

Size (favored by radars)

- Level 1 : Small, Medium and Large
- Level 2 : an approximate number of air units (from -25% to +50%)

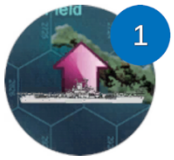
Models (favored by visual observation)

- Level 1 : dominant number of engines or mixed
- Level 2 : exact models but without info on carried payload

#7 – NAVAL MOVEMENT

Naval Move

1. Activate Map View in the menu



Each ACTION PHASE, you may move a fraction of your Task Forces
Allowed number is shown on the naval move icon in the menu
(If you don't have at least 4 TF, you won't see a number every Phase)

The number of allowed moves this PHASE is shown in the blue badge. On the last PHASE of the TURN, this number is displayed in a yellow badge for TF on autopilot, or in a red badge for other TF

2. Select a TF to move

4. DRAG & DROP

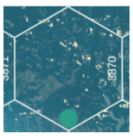
5. The TF is marked as moved
It will be able to move again NEXT TURN

3. Valid destination hexes are highlighted

Naval Move Restrictions

No move into or through ground hexes

No move through reef hexsides



Carriers may never move into coastal hexes

Speed

A Task Force speed is equal to the slowest speed among its ships

Crippled — The ship can only move one hex every 6 TURNS (towed)

Very Slow — May move only 1 hex every 3 TURNS
It moves only in the 06:40 to 07:40 TURN, in the 10:40 to 11:40 TURN, etc...

Slow — May move only 2 hexes over 3 TURNS
It cannot move in the 06:40 to 07:40 TURN, in the 10:40 to 11:40 TURN, etc...

Normal — May move one hex every TURN

Fast — May move up to 4 hexes over 3 TURNS
It moves 2 hexes in the 06:40 to 07:40 TURN, in the 10:40 to 11:40 TURN, etc...

Very Fast — May move up to 5 hexes over 3 TURNS
It moves 2 hexes except in the 06:40 to 07:40 TURN, in the 10:40 to 11:40 TURN, etc...

TF movement and status

Ready — The Task Force has not yet moved this TURN

Moved — The TF has already moved this TURN and cannot move again until next TURN

Retiring — The TF retires after complete unload or after heavy damage
It will automatically move at the beginning of the TURN

ASW — The TF is busy depth-charging an enemy sub and will not move until ASW operations are over

Autopilot is ON — A TF with autopilot will automatically move toward its programmed destination at the end of the TURN

Undo naval move

You may take back the naval move of any owned Task Force, except:

- If the TF has just entered an enemy-occupied hex
- If the TF has just launched an airstrike after moving
- If the TF has merged after moving
- If the current player action PHASE is over

Autopilot

You may activate autopilot for any of your Task Forces:

- You may still move it normally
- If it has not moved at the end of the TURN, it moves automatically towards its destination hex (emphasized in yellow)
- Autopilot is deactivated when the TF reaches its destination
- At any time, you may switch the autopilot off

Engagement

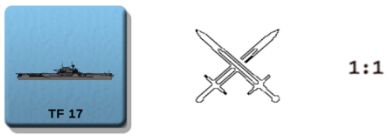
When moving into an enemy-occupied hex, your Task Forces WILL attempt to engage the opponent

Enemy reaction may vary:

- Fight
- Evade combat after a fight at longer range which means less damage
- Avoid combat and retreat
  If there are other enemy TF in the hex, another attempt is made against one of them
- Vanish from the map because its last known location was wrong

Engagement Ratio

In case of enemy naval attack on this Task Force or when this TF uses autopilot to move in an enemy-occupied hex



- Select a ratio
- If the ratio between the surface factors of your TF and the surface factors of the opposing TF is equal to or greater than the chosen ratio, then your TF will accept the combat
- Otherwise, it will attempt to escape
- This does not apply to the attacks resulting from your regular naval moves

Escape attempt

To check that a Task Force succeeds in escaping, a die-roll is made

- +3 if the TF sails faster than its opponent
- -3 if the TF sails slower than its opponent
- -5 if the TF is crippled
- +1 if the opposing TF has been detected

Results based on modified die-roll:

5+	Successful escape. The TF retreats to an adjacent hex
1-4	The TF is compelled to fight but at long range
< 1	The TF must fight

Fuel & Ports

In a few scenarios, TF may use full speed and get a limited fuel supply

- Each time they move by 2 hexes, they expend 4 fuel points
- Each time they move by 1 hex, they expend 1 fuel point
- If they run out of fuel points, they must move slowly
- The TF may only merge in a friendly port

Each turn spent in a friendly port

- The TF is supplied with 20 fuel points with a maximum of 60
- Torpedoes are fully reloaded
- Ships may again perform naval bombardment

#8 – SURFACE BATTLES

Surprise

Before the battle starts, each side rolls a die
Apply the following modifiers

- +2 for the Japanese
  (tactics and long-range torpedoes)
- +1 for the Japanese at night
- +1 for the US in 1943
- +1 with surface radar
- +1 if the enemy Task Force was detected prior to the battle

If a side has at least 4 more points (2 only at night) than its opponent, it achieves surprise

Torpedoes

The side with the surprise has the following benefits

- each of its ships equipped with torpedoes launches 1 torpedo salvo
  2 salvos may be fired at night with at least 4 surprise points
- each fired salvo expands 1 torpedo ammunition
- damage is assigned to the enemy ships before the gunfire step
- these torpedo-firing ships will still use their regular guns to fire at the enemy ships

Each salvo has a 25% chance to inflict one point of damage to a random target ship

- + 10 against slow ship
- + 20 against a crippled ship
- -5 against a very fast ship
- -5 against a DD
- + 10 at night
- - 10 if either side is evading
- - 20 for US torpedoes before 1944 (often dud)
- - 5 with at least 1 disorder point

Gunfire

Each surface ship fires with a strength equal to its surface combat factor

Each point of strength has a 25% chance to hit the target

- + 10 with fire control
- + 10 against a crippled ship
- + 10 against a burning ship at night
- - 5 for a DD against a CL, CL vs CA or CA vs BB
- - 10 for a DD against a CA or CL vs BB
- - 15 for a DD against a BB
- - 10 if either side evades
- - 5 rain
- - 10 light storm
- - 5 with at least 1 disorder point

Targeting

When using gunfire, each point of surface factors is assigned to a random target

However, the class of firing and target ships has an influence on the distribution as shown by the below table

Target class

Firing class	BB	CA	CL/DD	TRS/CV
BB	7	7	1	3
CA	3	3	3	1
CL/DD	3	3	7	1

Chance to fire upon

Example: a BB is seven times as likely to fire at another BB as at a DD

Outcome

The defending Task Force is beaten and must leave the hex if:

- it has opted to evade the battle OR
- it has more sunk, burning, or crippled ships than the attacking TF

Fletcher

Denver

Chicago

Otherwise, the attacking TF must retreat

Note: When a defender opts to evade battle. It is attempting to flee the hex before combat occurs. If successful, defender leaves and avoids combat. However, sometime this attempt is unsuccessful, and the defender is forced into battle. Regardless of outcome the defender must still leave the hex

Disorder

Each Task Force can get up to 8 points of disorder

- +1 for fighting at less than 2:1 in number of ships
- +1 per surface battle
- +2 for being surprised by the other side
- +2 for losing the battle without evading
- +1 if the roll of 1d100 is below the % of its ships which have been hit at least once during this battle (2 rolls)
- +1 if the roll of 1d100 is below the % of its ships which are sunk, burning or crippled (2 rolls)
- -1 the whole enemy TF has been sunk
- +4 when performing shore bombardment

A TF with at least 1 point of disorder has the following effects on the TF for a surface battle:

- -1 to the surprise die-roll
- -5 to the torpedo/gunfire of each of its ships

Disorder

Each Task Force can get up to 8 points of disorder

A TF with at least 1 point of disorder:

- may not voluntarily move into a hex containing enemy TF
- may not merge with another TF
- will fight a surface battle less efficiently
- may not bombard

The disorder is reduced by 1 point at the end of each TURN

#9 – UNLOADING AND BOMBARDMENT

Unloading Transports

- If it has spent the full TURN in its objective hex
- With a maximum of 2 cargo points per ship
- With a maximum of 6 points per enemy-controlled objective (invasion) and 12 points per friendly-controlled objective (reinforcements)
  Halved in a storm hex
- If target hex is clear of any enemy naval TF

A transport Task Force loads/unloads its cargo at end of TURN:

Some airfields have a limited fuel supply
Each unloaded cargo point supplies the airbase with 6 fuel points

Nowaki

The US and especially the Japanese also use DD to transport troops
(The famous Tokyo Express)

After complete unloading, a Japanese TF retires to Rabaul or to the north map edge. US transports retire to the south, escorted by at least half an equivalent number of DD (or CL/CA/BB if not enough)

Naval Bombardment

Each BB bombards with a strength equal to 100% of its surface combat factor, CA 80%, CL 60%, and DD 40%

- Each point of strength has a 25% chance to hit
  -5 rain
  -10 light storm
  +5 for every 8 grounded air units

- Each 2 hits (1 for a BB) inflict one point of damage to an enemy airfield and disable one coastal gun if any

Guadalcanal

- Each hit aborts, damages or destroys one random grounded air unit

- Against a beachhead, each hit reduces the number of unloaded cargos by 1 point

Naval Bombardment

A surface Task Force may bombard:

- an enemy airfield hex
- an enemy beachhead hex
  (an objective being invaded by the opposite side during this scenario)

A surface TF will bombard at end of TURN:

- If it has spent the full TURN in the target hex
- If target hex is clear of any enemy naval TF
- It has no disorder points
- Each ship may only bombard once a SCENARIO

Coastal defense

At the end of each TURN, the costal defense fires on each TF in its hex. Its surface fire like a DD

Hit TF can be forced to withdraw from hex if:

- It contains neither BB or CA
- The result of 1D10 is lower than twice the number of hits inflicted on the TF

Hence, it is important to reduce the defense by air or sea before invading

#10 - AIR ATTACKS

Anti-Aircraft Fire

Heavy AA fires further and shoots down enemy aircraft BEFORE they release their bombs or torpedoes

- Its value is printed on the counter of the airfields
- For a US TF, it is equal to the sum of the best 4 heavy AA of the ships and cannot exceed 100. A Japanese TF only sums up its 2 best ships. Double the number of firing ships against an airstrike detected before the battle. When a TF and a friendly base share the same hex, sum their heavy AA

Light AA destroys enemy planes only AFTER they release their bombs or torpedoes

- Its value is printed on the attacked airfield or individual ship
- Strategic Bombers are not sensitive to light AA because they are flying too high
- Doubled against aircraft carrying torpedoes
- Halved if both torpedo- and dive-bombers are attacking
- The light AA of each US carrier is increased by the light AA of the best operative escort ship (share best escorts between several carriers)

Each AA point has a 4% chance to hit an enemy bomber or fighter-bomber

- 3% only with rain
- 2% only with light storm

Anti-Aircraft Fire

Heavy AA fires further and shoots down enemy aircraft BEFORE they release their bombs or torpedoes

- Its value is printed on the counter of the airfields
- For a US TF, it is equal to the sum of the best 4 heavy AA of the ships and cannot exceed 100. A Japanese TF only sums up its 2 best ships. Double the number of firing ships against an airstrike detected before the battle. When a TF and a friendly base share the same hex, sum their heavy AA

Light AA destroys enemy planes only AFTER they release their bombs or torpedoes

- Its value is printed on the attacked airfield or individual ship
- Strategic Bombers are not sensitive to light AA because they are flying too high
- Doubled against aircraft carrying torpedoes
- Halved if both torpedo- and dive-bombers are attacking
- The light AA of each US carrier is increased by the light AA of the best operative escort ship (share best escorts between several carriers)

Each AA point has a 4% chance to hit an enemy bomber or fighter-bomber

- 3% only with rain
- 2% only with light storm
- +1 Allied heavy AA in 43, +2 in 44.

Air-to-Ground Attack

Each air-to-ground factor of surviving air units has a 20% chance to hit grounded aircraft

- +5 if the airfield has no CAP
- +1 for each grounded unit being prepared for a mission or which has just landed outside
- -1 for each 5 pts of AA
- There is always a minimum 5%-chance to hit
- -5 strategic bombers by overcast weather
- -5 rain
- -15 light storm
- Each hit destroys 1 grounded air unit
- -5 if the bomber is confused

Each air-to-ground factor of surviving air units has the same chance to hit the runways

- Fighters without bombs (strafing) cannot hit runways

#11 - AIR COMBAT

Air Combat Procedure

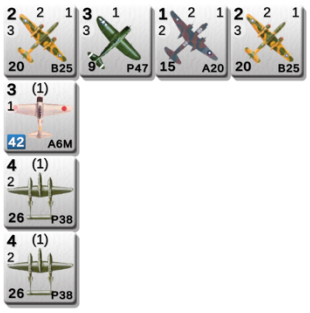
The planes are split into stacks

- Attacking bombers and fighter-bombers are placed each in one stack
- Then defending fighters are evenly and randomly placed in these stacks
- Then escorting attacking fighters are aligned evenly and randomly against the defending fighters

A single round of air combat is fought within each stack

Examples

2xB25, 1xA20, 1xP47 as fighter-bomber and escorted by 2xP38 vs. 1xA6M on CAP



2xB25, 1xA20, 1xP47 as fighter-bomber and escorted by 2xP38 vs. 2xA6M, 1xKi43, 1xKi61 on CAP

Air to Air Table

Both sides roll simultaneously using the column determined below:

- Best air-to-air value of its fighters +1 for each extra friendly fighter (or +2 without enemy fighter)
- Or air-to-air value of the unescorted bomber
- Then subtract the air value of the other side using the same rules

-1 rain

-2 light storm

Example: The Japanese (air value of 3) fire at -2 while the US (air value of 5) fire at +2

<-3	-3	-2	-1	0	+1	+2	+3	+4	+5	+6	7+
5%	5%	10%	15%	20%	20%	20%	20%	20%	20%	20%	25%
5%	10%	10%	10%	20%	20%	25%	25%	25%	30%	35%	40%
						5%	10%	15%	20%	25%	25%
5%	10%	15%	20%	20%	25%	20%	20%	20%	15%	10%	5%
85%	75%	65%	55%	45%	35%	30%	25%	20%	15%	10%	5%

- 1 enemy air unit is hit, if possible, a bomber or fighter-bomber
  If rolled by a bomber without escort, then the result is treated as no effect instead
- 1 enemy air unit is hit, if possible, a fighter
- 2 enemy air units are hit
- Enemy bomber gets confused: it attacks a random ship, and with a penalty
- No effect

CAP - Ammunition

A CAP unit has a 50% chance to be ammunition-depleted after combat
This fluctuates with the ratio CAP units against attacking air units

25% for a US unit with a ratio > 2:1

75% for a US unit with a ratio < 1:2

25% for a Japanese unit with a ratio > 3:1

75% for a Japanese unit with a ratio < 1:1

Japanese fighters embed less ammo than their Allied counterparts

Improvised CAP

Bombers SBD and D4Y may be allocated to CAP missions, although they are less efficient than pure fighters

When firing at opposing air units, its air-to-air factor is reduced by 2
This is emphasized by the parenthesis around the air-to-air rating

Ex: the SBD fires on the 0 column and the G4M fires back on the -2 column

When defending with a friendly fighter, they provided a +1 only against an unescorted bomber

CAP Vectoring

Operative carriers with air radar have a chance of vectoring CAP planes toward incoming enemy air units.

10%

10% before 43, 30% in 43, 50% in 44.

30% before 43, 40% in 43, 50% in 44.

-5% if both torpedo and dive bombars are attacking.

-20% if the airstrike has not been detected.

If 1D100 is less or equal than the CAP vectoring chance, then the CAP will fight two rounds of air combat BEFORE the bombers attack.

Otherwise, then the CAP will fight the second rounf of air combat AFTER the bombing.

#12 – DAMAGE AND REPAIR

Air Unit Damage

When an air unit is hit, it may be destroyed, damaged, or aborted depending on its ruggedness

	Fragile	Standard	Rugged
	20 %		40 %
	30 %		40 %
	50 %		20 %

A damaged or aborted air unit must immediately abort its mission and jettison its payload

A damaged unit in a hangar has a 6% chance each phase to be repaired (3% if Japanese)

A damaged unit may not be allocated any mission until repaired

An aborted unit must just spend an extra PHASE in the hangar in order to be operative again

See ships damage

See the damage report for the ships of the Task Force

Major fire

The flight deck is not usable

Crippled

Sunk when all life points have been lost

Heavy Damage

A surface ship is heavily damaged when:

- she has lost half of her life points
- OR she has lost her propulsion
- OR she has lost half of her main turrets
- OR she is facing a major fire

Such a ship must retire with a small escort to Rabaul or the north map edge (Japanese) or to the south (US)

A carrier TF left without escort will be forced to withdraw

Getting damaged

By a torpedo
Loss of 1 life point
On a failed Damage Control Test (DCT) (1D10 > damage control), the carrier lists
On a failed DCT, the propulsion gets damaged
1 potential critical hit

By a bomb
Up to 30% of AA guns destroyed
75% chance to damage the flight deck, 10% to destroy a Japanese flight deck, 1% if US
1 potential critical hit

By a shell
Up to 30% of AA guns destroyed
1 potential critical hit

Extra effects on carriers
A hit destroys 2 air units
BUT if 2 armed units (in the middle of their preparation for their mission) are hit, then more planes are destroyed, a major fire breaks out and there is 1 potential critical hit for each 2 bombers destroyed this way

Airfields
Each hit on an airfield reduces its life points by 1
For each 5 life points lost, one runway is made inoperative
Life points may go below zero

Damage to propulsion:
Each time the propulsion gets hit, compare the cumulative number of hits to the ship's current life points
If greater, then the propulsion is destroyed
If tied, then a failed DCT cripples the ship

Critical hits

For each potential critical hit, if the result of a die-roll is greater than the damage control value of the ship, then the ship receives one critical hit from the list below
Carriers attacked by a detected airstrike have +1 to their damage control value

Damaged flight deck
There are holes in the flight deck preventing any landing and launch

Destroyed flight deck
The flight deck has been wrecked. No aircraft can land on or take off from the carrier for the rest of the game

Major fire
If not under control at the end of turn, then the fire will cause more critical damage. No air or repair operations allowed on a burning ship except firefighting

Minor fire
Extinguished automatically during next phase. 2 minor fires in the same phase trigger a major fire

Scuttled due to uncontrolled fires
The crew had to abandon the ship and to scuttle it to prevent its capture

Damaged propulsion
The ship will move slower

Propulsion destroyed
The ship is dead in the water for the remainder of the game and cannot land or launch air units

1 elevator destroyed
Air operations are now slower

Reduced damage control capability
The ship damage control value is halved, hampering damage prevention and repair operations

Critical hits

1 main turret destroyed
The surface battle value of the ship is reduced

Disabled torpedo launchers
The ship will not launch any torpedo in a surface combat

Ammunition explosion
Loss of 1 life point and a major fire breaks out. Carrier airplanes may not be equipped with a full load of bombs anymore

Avgas explosion
Loss of 1 life point and a major fire breaks out

Torpedoes explosion
Loss of 1 life point and a major fire breaks out. The ship may not use her torpedoes anymore for surface battles or on her planes

Oil leak
Enemy scout planes have more chance to find this TF

Structural damage
The ship loses 1 life point

List due to flooding
No air operations allowed on a carrier which lists

Flooding not controlled
No measures could stop the water to enter the ship en masse which then sank inexorably

Air radar destroyed
It cannot detect incoming enemy air strikes anymore. Enemy airstrike detection should rely on other ships of the Task Force

Fire control destroyed

Critical hits

Structural damage
The ship loses 1 life point

List due to flooding
No air operations allowed on a carrier which lists

Flooding not controlled
No measures could stop the water to enter the ship en masse which then sank inexorably

Air radar destroyed
It cannot detect incoming enemy air strikes anymore. Enemy airstrike detection should rely on other ships of the Task Force

Fire control destroyed
The radar-assisted gunfire system is down. The ship loses its bonus in surface battle

Surface radar destroyed
No effect if the TF contains other ships equipped with surface radar. Otherwise, this will increase the chance to be surprised by the enemy

Seaplanes capability lost
The search capability of the task force decreases

Service capacity on the deck destroyed
Arming and fueling CAP units cannot be done on the deck anymore but may be still performed in the hangar, which means using elevators and thus losing time

1 pt. of cargo destroyed
1 pt. of cargo has been lost

Reduced unloading capacity
This ship may only unload 1 pt. of cargo a TURN

Repair

Saratoga

Damaged flight deck
One repair attempt at the end of each PHASE. Repaired if a die roll is less than or equal to the ship damage control value

San Juan

Damaged propulsion
One repair attempt at the end of each TURN
Under control if a die roll is less than or equal to the ship damage control value

Princeton

List due to flooding
One repair attempt at the end of each TURN
Under control if a die roll is less than or equal to the ship damage control value

San Diego

Major fire
One repair attempt at the end of each TURN. Under control if 2 die rolls are each less than or equal to the ship damage control value. If one fails, the ship gets 1 potential critical hit. If both fail, the ship is abandoned by its crew and scuttled

Midway

Airfields
One repair attempt at the end of each PHASE. Heal 1 life point if a die roll is less than or equal to the airfield damage control value (2)

Armored Flight Deck

Formidable

In yellow: armored flight-deck

- The chance to get an inoperative flight deck following a bomb hit drops from 75% to 25% per hit.
- A bomb or shell has more chance to hit the deck area rather than the hangar area.
- Damage control is increased by 2 against a bomb hit, by 1 against a shell hit.

#13 - WEATHER

Weather

The weather comes an as an option in each scenario

Overcast

Rain

Light Storm

Clouds are moving 1 hex every 2 TURNS
75% in the wind direction shown on the cloud
25% at random

The wind is changing direction every 12 HOURS

When clouds stack with each other, rain and storm appear
Clouds leaving the map are randomly replaced

Detection

Chance of detection of Task Forces or approaching airstrikes are reduced when covered by weather. Radar is not affected by weather

- 50 %

- 65 %

- 80 %

Air Attacks

They are impacted by the weather in their hex

Strategic bombers suffer a -5 modifier to hit naval or ground targets
(Cumulative with other modifiers, see Air Attacks section)

Heavy AA hit chance is reduced to 3% per point
Bombers (except torpedo-bombers at sea) suffer a -5 modifier to hit naval or ground targets
Air units fire one column to the left on the Air Combat Table

Heavy AA hit chance is reduced to 2%, Light AA hit chance is reduced to 3%
Bombers suffer a -15 modifier to hit naval or ground targets
Air units fire two columns to the left on the Air Combat Table

Ships

Ships are impacted by the weather in their hex

No effect

Ships suffer a -5 modifier for gunfire and naval bombardment
(cumulative with other modifiers)

Ships suffer a -10 modifier for gunfire and naval bombardment
The unloading capacity is halved
Speed is reduced by one level (ex: slow -> very slow)

Air Units

Airborne air units are impacted by the weather in their hex

No effect

Landing on carriers is slightly slower
Radio -10

Airstrikes attempt to avoid storm hexes
If they enter a storm, they cannot move a second hex this PHASE
At the beginning of each PHASE, each airborne unit loses 1 point of endurance
Radio -20
Launch and landing on carriers are slightly slower

#14 – Multiplayer

Cross-platform multi-player

Play against anyone over internet on PC/Mac/iOS/Android

- Turn-by-turn asynchronous 2-player game. You don't need to both connected at the same time
- Take your time to play against a challenging human opponent
- The game state is automatically saved in the cloud
- Login and notifications by email
- Less cheat opportunity

You are limited to 10 online games
Finish and delete one before starting a new one

Join an online game

Go to saved games

Choose 2P games waiting for opponent

Sign up by email

Join an existing game if any. Otherwise, you may create a new one

Create an online game

Go to New Game and select a scenario

Check the 2P online checkbox

- By default you will play against the first player joining your game
- You may also search a player by nickname
- Or rematch a previous opponent

Pick up options, modules, and choose your side. Modules shall be activated only if the opponents plays them as well

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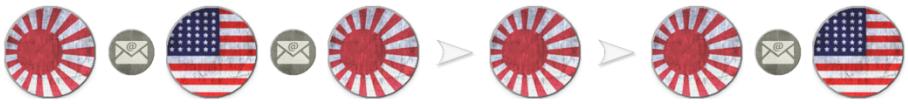
Sequence

Each PHASE, the Japanese player makes its moves and then hands over to his US opponent

If one player opts to skip one or more hours, then other player will play several PHASES in a row



If both players do so, this is good way to go faster into action
See the Sequence of Play section of the help

#15 – SEAPLANES - DLC

Manageable Seaplanes

By default, seaplanes are managed by the program. You can't see and interact with them

Getting the seaplanes extra feature will open more options:

- See all owned seaplane bases on map

- Assign search, CAP and transfer missions to your seaplane air units

- Establish temporary seaplane bases with seaplane tenders

When playing against AI, or against a player who purchased this feature too:

- You are also allowed to observe and attack enemy seaplane bases by air and by sea

SeaPlane Base

Maintenance Capacity (MC)

AA value (Heavy - Light)

May operate any kind of seaplanes

Coastal defense

The MC limits the number of seaplanes which could be operated per PHASE

- It increases by 1 for each friendly seaplane tender present in the hex

- It may be reduced by ground strike or shore bombardment, although it cannot go below zero
- 20% chance to repair by 1 point at end of each PHASE

SeaPlane Air Units

In White,
Large Seaplane

May operate only on shore seaplane bases

In Orange,
Small Seaplane

May operate on shore seaplane bases, and onboard ships with ORANGE hangar capacity

Select ship to enter Air Op

Temporary Seaplane Base

You may establish a temporary seaplane base with a TF if all below conditions are met:

- the TF has spent the full last TURN in a seaplane spot hex, orange & White (Also a coastal hex)
- The hex is free of enemy TF or base
- The TF contains at least one operative seaplane tender. Its carrierd air units are immediately moved to the base's hangar

The new base has zero maintenance capacity, and relies solely on tenders

It remains concealed to the enemy until the tender TF gets detected

#16 – ADVANCED FOG OF WAR – DLC

Advanced fog of war

This paid module offers more suspense as enemy losses, and the score remains uncertain until the end. You may get false reports on:

- Observed enemy ships' classes
- The damage inflicted on enemy air units

- The amount of hits inflicted by aircraft on enemy ships along with the resulting observed damage, including major fire, list, crippled, and sinking

The observed state of enemy naval units may only be updated:

- On a new battle
- On a new detection
- At the end of the game

Enemy air losses outside combat (night, fuel, etc...) remains concealed until the end of the game

Accuracy

Factors inducing inaccurate reports:

- CAP & AA Defensive Fire
- Bad weather
- Confused bombers

- Japanese observers

Factors favoring accuracy:

- Shadowed TF

- Previously observed ships
- Air-to-sea combat
- Surviving bombers

#17 – SUBMARINES – DLC

Submarines

Class
Submarines are not individually named in the game

Low maximal depth

Submarines have neither life points nor damage control, they are presumably sunk as soon as hit

Patrol

At the start of most scenarios, choose a patrol area and a mission for your submarines

Afterwards, you won't see your own subs until the end of the game.
Submarines will move automatically and randomly within 4 hexes of their start location, avoiding enemy airbases

Each sides also get 3 reserve submarines, which could appear anywhere on map, thus denying any safe haven for the enemy

ASW units

An ASW ship is a ship with any depth charges (check the back of the counter), usually a DD

An ASW air unit is a bomber placed in the ASW box of a TF

A scout detecting a submarine is also counted as ASW unit for the encounter with the sub
The chance to detect submarines is divided by 2 in clear weather, by 8 otherwise

ASW units are crucial to detect, evade and kill enemy submarines

Submarine encounter

When the sub moves, there is a encounter with an enemy TF in the same hex

If the TF is adjacent, the probability depends on weather and speed

- 70/50/30/10% in clear/overcast/rain/storm
- +20 for each point of difference between the sub's and the TF's speed

Otherwise, there is always a 1% chance of a random encounter of a reserve sub with any TF on the board each TURN

A scout air unit which detects the submarine also triggers an encounter

Surprise

Each side roll 1D10 and applies modifiers

For the submarine

- +2 against a TF in a coastal hex
- +2 against a crippled, loading or unloading TF
- +1 against a slow TF. -1 against a fast TF
- +2 against a TF without any ASW unit

For the escort

- +1 for the US in 43 (+2 in 44)
- +1 with surface radar at night
- +1 with at least 3 Japanese ASW ships
- +1/+2/+3 for the US with at least 1/3/6 ASW ships
- +1 with an ASW air unit. +1 in clear weather

Surprise

The side with the highest roll gets a number of surprise points equal to the difference which is reduced by 1 in storm and against a reco sub
Surprise points are spent in this order of priority

For the submarine

- 4 at night: one torpedo salvo at short-range against the TF and the sub escapes unscathed
- 4 at day for a reco sub: one attack at long range and the sub escapes
- 2: one torpedo salvo at short-range, then the submarine gets detected and forced to submerged
- 1 attack sub: 2 torpedo salvoes instead of 1 (bow + stern)
- 1: higher intelligence report

For the escort

- 4: surprise attack on the sub by a ASW units, then the subs is detected and dives
  If there is at least one ASW ship, she has a 10% chance to hit the submarine by guns or ramming
  If there is at least one ASW air unit, it has a 20% chance to hit the submarine by bombing or strafing
- 2: the submarine is detected and forced to submerged
- 1: lower intelligence report

Torpedo Salvo

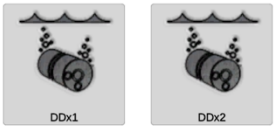
Each torpedo salvo fired by a submarine has a 25% chance to hit

- +20 at short-range
- +10/-20 Japanese/US submarine (before 44)
- +20/+10/-5 against a ship which is crippled/slow/very fast
- +10 at night
- -10 against a short ship (DD)
- -5/-10 rain/storm
- There is always a minimum 5%-chance to hit

Depth Charges

Each PHASE, a TF containing at least one ASW ship automatically depth charges a detected sub in its hex, provided its depth charge settings is different than None

Depending on the setting, 1 or 2 ASW ships may be detached for this duty



The chance to hit the submarine equals 5%

- +5 in coastal waters or against low-depth submarine
- +5 for the US

The process expands one depth charge per PHASE and is repeated until the submarine is sunk or undetected.
The sub may not move and attack during that time

Keeping contact

Depth charges detonations disturbs sonars. There is a 50% chance to keep contact with an enemy submarine after each depth-charge attack

- -20 if Japanese (doctrine, training and material)
- +10 US in 43
- +20 if there are at least 2 ASW ships (one keeps sonar acquisition while the other throws depth-charges

If contact is lost, then the submarine becomes immediately undetected

Contact is also lost when:

- ASW ships run out of ammunition
- ASW ships leave the hex

Location report

If the submarine survives an encounter with a TF, it will transmit the location of the TF
The location is approximative with an Attack mission

Due to the surprise, the report may be lower

- Approximative for a Reco sub
- Absence of report for an Attack sub

With a higher report, the location is always accurate

#18 – NIGHT AIR OPERATIONS

Night Air Op.

Air units with embedded radar may carry air operations at night.

Only British CV air units have this capability (See Indian Ocean scenarios).

All the following rules apply only to them.

Weather is shifted one step toward Clear (Ex: Rain -> Overcast).

Intelligence increase -4 to the die-roll instead of -2.

Detection of the scout: -1 to the die-roll.

There is no CAP and no AA fire.

The chance to hit a ship is reduced by 15 pts.

No crash on landing.

#19 – INTO THE WIND - DLC

Introduction

With the obviously very limited length of the flight deck, taking off required a good wind over the deck. The carrier would steam into the wind at a good speed, adding the wind generated by the ship displacement, to produce a good wind over the deck.

Wind direction

The wind direction is represented by a small arrow at the bottom of each hex.

Deck Length

Each carrier has its own deck length.
It is visible on the flipped ship counter.
The length limits the number of air units which may be launched in one PHASE.

Air Operations Points

At the end of each PHASE, for each aircraft carrier, air op pts are collected depending on flight deck activity

- If the number of air units which landed or launched this phase (N) is below 2, this step is skipped.
  Idem with only 2 air units landed.
- Otherwise, the number of points increases by 1.
- A second point is obtained with a modified X-sided die is inferior or equal to N, X being the carrier deck length.

Modifiers

- -1 for a full deck-load (less deck length for the most forward planes)
- -2 if at least 2 torpedo bomber units have been launched (heavy payload)
- -1 at night
- -1 in stormy weather
- -1 if the carrier task force contains a ship (carrier or escort) with a maximum speed < 5
- -1 same slow task force in Clear weather (not enough wind over the deck)
- +1 for AI

The amount of points is displayed next to the windsock.

Air op pts are cumulated during the whole TURN and reset at the beginning of each TURN.

Example:

Enterprise launches 3 air units in CAP and 2 in SEARCH this phase. She is occupying a storm hex.
Enterprise gains 1 air operation point.
Her deck has a length of 8. A D8 is rolled at +1 (storm) and gives a 4 modified to 5. Hence a second point is recorded.

Relative sailing direction compared to the wind

When the carrier moves in the opposite direction of the wind, it is sailing into the wind.
Let's borrow terms from organic chemistry (benzyne ring) to name the other directions.

- PARA : Opposite direction of the wind (into the wind)
- META : Next to the opposite direction
- ORTHO : Next to the same direction
- SAME : Same direction as the wind

Carrier Air Operation effects on Naval Movement

As air operations required the carrier to stream into the wind, moves in the wind direction are the most penalized.

The following table gives the speed increase / decrease depending on the relative direction to the wind and the number of air operation points.

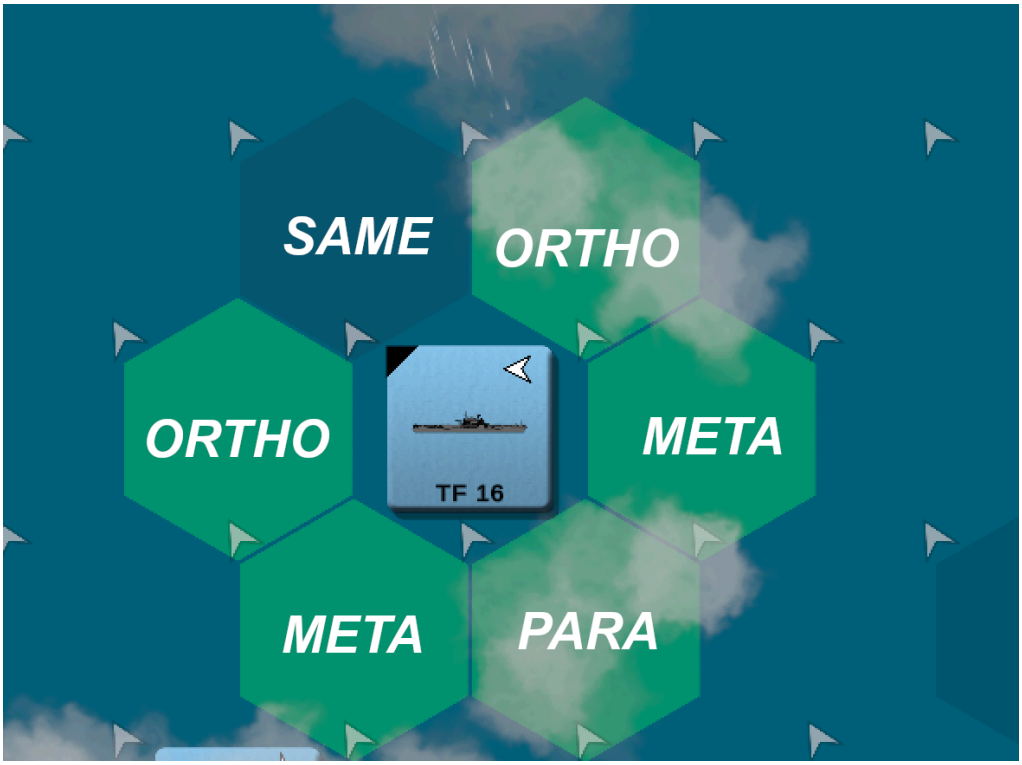
Direction / Air op pts	2	3	4	5	6-7	8+
SAME	-1	-1	-2	-2	-3	-4
ORTHO	0	-1	-1	-2	-2	-3
META	0	0	+1	+1	+1	+2
PARA	0	+1	+1	+1	+2	+2

Speed is always limited by the maximum speed of the slowest ship in the task force.
The maximum speed is displayed on the back-side of the ship counter.

Hence, depending on the wind, movement conditions and air operations, some move directions might not be available, and some direction might allow a 2-hexes move.

Example

Enterprise movement potential with 4 air op pts. Suppose she has moved last turn.
There is not enough speed left (3-2 = 1) to move in the wind direction (NW) this turn.
With 6 pts, Enterprise could have moved 2 hexes in the SE direction (speed 3+2 = 5).



Reminder on naval speed

- Speed 3: a 1-hex move each TURN
- Speed 2: a 1 hex-move only if the TF has not moved twice the last 3 TURNS (up to 2 hexes every 3 TURNS)
- Speed 1: a 1-hex move only if the TF has not moved the last 3 TURNS (up to 1 hex every 3 TURNS)
- Speed 4: a 2-hexes move only if the TF has not moved the last 3 TURNS, one hex otherwise (up to 4 hexes every 3 TURNS)
- Speed 5: a 2-hexes move only if the TF has not moved twice the last 3 TURNS, , one hex otherwise (up to 5 hexes every 3 TURNS)

Naval Movement effects on Carrier Air Operations

Both landing and take-off are impacted by the carrier general movement direction during the turn.

The wind socket adopts a color code to reflect this impact in the air operation screen.

Wind socket	Relative direction	Effect on the maximum number of air units which may land / takeoff per PHASE
	NONE	No effect
	SAME	1 (take-off) or 2 (landing)
	ORTHO	Half deck-length (rounding up for AI, down otherwise)
	META	Deck-length
	PARA	Deck-length + 1

Example:
If Enterprise has moved next to the wind direction (ORTHO), she could launch/land only up to 4 air units (half of the deck length) per phase.

#20 – PILOTS - DLC

Introduction

Japan had expected a short conflict with Occidental powers. It had skillful elite pilots but very few training programs.
As the war protracted, the overall pilot quality quickly start to worsen, especially after the attritional battles of Midway and Guadalcanal.

Inexperience

The % if inexperience is determined by the Japanese side only.

- 20% end 1942
- 30% in 1943
- 40% in 1944
- +10% for land-based air units from 1943.

Test & Penalty

On specific actions, an experience test is done on an air unit, a D100 is rolled.
If the result is less or equal than the inexperience level, then a penalty is applied to that test.

Action	Penalty on failure
Air-to-air combat, start	Air-to-air value - 1
Air-to-air combat, end	Confused
Air-to-sea combat	Does not count toward coordination bonus
Search	Chance -10
Scout interception	-1 Japanese, +1 US
Intelligence upgrade	-1
Landing with lack of endurance	Crash chance +10

#21 WARNING

Please Read Before Using This Game Or Allowing Your Children To Use It.

Some people are susceptible to epileptic seizures or loss of consciousness when exposed to certain flashing lights or light patterns in everyday life.

Such people may have a seizure while watching television images or playing certain video games. This may happen even if the person has no medical history of epilepsy or has never had any epileptic seizures.

If you or anyone in your family has ever had symptoms related to epilepsy (seizures or loss of consciousness) when exposed to flashing lights, consult your doctor prior to playing. We advise that parents should monitor the use of video games by their children. If you or your child experience any of the following symptoms: dizziness, blurred vision, eye or muscle twitches, loss of consciousness, disorientation, any involuntary movement or convulsion, while playing a video game, IMMEDIATELY discontinue use and consult your doctor.

- Do not stand too close to the screen. Sit a good distance away from the screen, as far away as the length of the cable allows.
- Preferably play the game on a small screen.
- Avoid playing if you are tired or have not had much sleep.
- Make sure that the room in which you are playing is well lit.
- Rest for at least 10 to 15 minutes per hour while playing a video game.

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#22 – SUPPORT

All questions related to technical issues with the game CARRIER BATTLES 4 GUADALCANAL can be sent by email to the support of SAS AVALON DIGITAL at the following email address:

support@avalon-digital.com or carrierbattles@yahoo.com